Interactivity and navigation

Worked example: the complete Stopwatch app (`code/week06/examples`)

Key ideas

- **UI** is *what* is on the screen; **UX** is *how* it reacts: interactivity, navigation, animation.
- Build only the visible rows of a long list.
- Replace a route when the user must not come back (login).
- Dialogs and modal bottom sheets are **routes**: pop closes them, and `pop(value)` answers.

ListView.builder for long lists

```
Scrollbar(
  controller: _scrollController,
  child: ListView.builder(
    controller: _scrollController,
    itemExtent: itemHeight,     // 60 px rows
    itemCount: _laps.length,
    itemBuilder: (context, index) =>
      ListTile(
        title: Text('Lap ${index + 1}'),
        trailing:
          Text(_secondsText(_laps[index]))),
  ),
)
```

`itemBuilder`	builds row `index` when it comes on screen
`itemCount`	**without it the list is infinite**: `RangeError`
`itemExtent`	fixed row height: no measuring; row n is at $n\times$ height
`controller`	a `ScrollController`

`Scrollbar` gets the **same** controller as the list.

ScrollController: move the list from code

```
final _scrollController = ScrollController();

void _scrollToEnd() {
  if (!_scrollController.hasClients) return;
  final visible = _scrollController
      .position.viewportDimension;
  final end =
      itemHeight * _laps.length - visible;
  if (end <= 0) return;   // all laps fit
  _scrollController.animateTo(end,
    duration: const Duration(milliseconds: 500),
    curve: Curves.easeIn);
}
```

```
@override
void dispose() {
  _timer?.cancel();
  _scrollController.dispose();
  super.dispose();
}
```

`animateTo` slides; `jumpTo(offset)` goes at once. A controller lets code *outside* `build` act on a widget.

Login screen details

One screen, two faces: a `bool` picks the tree.

```
body: Center(child: _loggedIn
    ? _buildSuccess() : _buildLoginForm()),
```

Email: `keyboardType: TextInputType.emailAddress` and

```
final regex =
  RegExp(r'^[^@\s]+@[^@\s]+\.[^@\s]+$');
if (!regex.hasMatch(text.trim()))
  return 'Enter a valid email';
```

`^ $`	start, end of the text (else: match anywhere)
`[^@\s]+`	one or more chars, not @ or space
`\.`	a real dot (`.` = any char)
`r'...'`	raw string: `\` is ordinary

`validate()` **is** `true` **when** ***all*** **fields are valid. Stop when it is** `false`**:** `if (!_formKey.currentState!.validate()) return;`**. The printed** `if (form?.validate() ?? false) return;` **logs in with an** ***invalid*** **form.**

Choose how to navigate

Method	Stack after
`pushReplacement(r)`	current route replaced by `r`: no way back
`pushAndRemoveUntil(r, test)`	pushes `r`, removes routes below until `test` is true; `(r) => false` removes all
`canPop()`	`true` if a route is below

```
// after a valid login
Navigator.of(context).pushReplacement(
  MaterialPageRoute(builder: (_) =>
    StopWatch(name: name, email: email)));
// log out: start from an empty stack
Navigator.of(context).pushAndRemoveUntil(
  MaterialPageRoute(builder: (_) =>
    const LoginScreen()),
  (route) => false);
```

- The State reads constructor data through `widget`: `Text(widget.name)`.
- The AppBar shows a back arrow only when `canPop()`.
- Unused builder parameter: name it `_`.
- Transition: slide on iOS; fade and zoom (predictive back) on Android.

Dialogs

```
showDialog<void>(
  context: context,
  builder: (context) => AlertDialog(
    title: const Text('Run completed!'),
    content: const Text('Total: 12.3 s'),
    actions: [TextButton(
      onPressed: () =>
          Navigator.of(context).pop(),
      child: const Text('Close'))],
  ),
);
```

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Material	Cupertino
showDialog	showCupertinoDialog
AlertDialog	CupertinoAlertDialog
TextButton	CupertinoDialogAction
tap outside closes	tap outside does nothing

Show a dialog only when the user must react: it interrupts. `barrierDismissible: false` blocks the outside tap.

Platform-aware dialog

```
Future<void> show(BuildContext context) {
  final p = Theme.of(context).platform;
  if (p == TargetPlatform.iOS ||
      p == TargetPlatform.macOS) {
    return _showCupertinoAlert(context);
  }
  return _showMaterialAlert(context);
}
PlatformAlert(title: 'Run completed!',
  message: 'Total run time is ...')
  .show(context);
```

Built in: `showAdaptiveDialog` + `AlertDialog.adaptive`. Test iOS anywhere with `ThemeData(platform: TargetPlatform.iOS)`. `fold(start, f)` sums the laps: `_laps.fold(_ms, (t, l) => t + l)`.

Wait for the answer

```
final ok = await showDialog<bool>(
  context: context,
  builder: (context) => AlertDialog(
    title: const Text('Clear all laps?'),
    actions: [
      TextButton(onPressed: () =>
        Navigator.of(context).pop(false),
        child: const Text('Cancel')),
      TextButton(onPressed: () =>
        Navigator.of(context).pop(true),
        child: const Text('Clear')),
    ]));
if (ok != true || !mounted) return;
setState(() => _laps.clear());
```

- The result is `T?`: `null` after a tap outside or Back. `if (!ok)` does not compile on a `bool?`.
- After `await`: `if (!mounted) return;` (`context.mounted` in a stateless widget).

```
final m = await showDialog<int>(
  context: context,
  builder: (context) => SimpleDialog(
    title: const Text('Distance of one lap'),
    children: [for (final d in [100, 200, 400])
      SimpleDialogOption(
        onPressed: () =>
            Navigator.of(context).pop(d),
        child: Text('$d m'))]));
```

Bottom sheets

Modal	Persistent
showModalBottomSheet	showBottomSheet of Scaffold.of
page darkened, blocked	page still works
a route: pop(value)	part of the Scaffold
Future<T?>	a controller: close(), closed

```
final chosen =
  await showModalBottomSheet<LapOrder>(
    context: context,
    builder: (context) => SafeArea(
      child: Column(
        mainAxisSize: MainAxisSize.min,
        children: [for (final o in LapOrder.values)
          ListTile(title: Text(o.label),
            onTap: () =>
              Navigator.of(context).pop(o))])));
if (chosen == null || !mounted) return;
```

`MainAxisSize.min`: sheet only as tall as its content. `SafeArea`: away from system bars.

```
bottomNavigationBar: Builder(  // below Scaffold
  builder: (scaffoldContext) => ElevatedButton(
    onPressed: () => _showSummary(scaffoldContext),
    child: const Text('Summary'))),

final c = Scaffold.of(scaffoldContext)
    .showBottomSheet((context) => summary);
_closeTimer = Timer(
    const Duration(seconds: 5), c.close);
c.closed.then((_) => _closeTimer?.cancel());
```

The complete Stopwatch app

main.dart	home: the login screen
login_screen.dart	Form; valid → pushReplacement with name, email
stopwatch.dart	laps list, alert on Stop, sheet on a lap tap, Log out with confirmation
platform_alert.dart	PlatformAlert

Two files may import each other (login ↔ stopwatch).

Old API in the printed text

headline5	headlineSmall
styleFrom(primary:)	backgroundColor, foregroundColor
MaterialStateProperty	WidgetStateProperty (or styleFrom)
Theme.primaryColor + white	colorScheme.primary + onPrimary
CupertinoButton in dialog	CupertinoDialogAction
validateAndSubmit	no such method: validate, save, reset
late Timer timer	Timer? _timer, _timer?.cancel()

Common errors

RangeError while scrolling	add itemCount
ScrollController used after being disposed	dispose once, in dispose
Back returns to login	pushReplacement
Dialog never closes	its button must pop
Scaffold.of() ... does not contain a Scaffold	a Builder below the Scaffold
A nullable expression can't be used as a condition	ok != true
BuildContexts across async gaps	if (!mounted) return;
Material dialog on iPhone	PlatformAlert or .adaptive

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